

AP CALCULUS AB - UNIT 4

Guided Notes 4.5: Linear Approximation



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we construct the linearization of a differentiable function at a convenient input?

- Construct the linearization of a differentiable function at a convenient input.
- Approximate nearby values using a tangent line.
- Use differentials to estimate changes and propagated error.
- Use concavity to determine whether a linear approximation overestimates or underestimates.

Key Knowledge, Vocabulary, and Notation

Linearization

$L(x) = f(a) + f'(a)(x - a)$, the tangent line at $x = a$.

Local approximation

For x near a , $f(x) \approx L(x)$.

Differential

$dy = f'(x) dx$ approximates $\Delta y = f(x + \Delta x) - f(x)$.

Error sign

If $f'' > 0$, tangent lines lie below the curve locally; if $f'' < 0$, they lie above.

Absolute error

$|\text{actual} - \text{approximation}|$.

Relative error

Absolute error divided by the magnitude of the actual value, often expressed as a percent.

Concept Development and Strategy

1. Choose a base point a where both $f(a)$ and $f'(a)$ are easy to compute.
2. Write the tangent line first; then substitute the nearby target input.
3. Keep exact arithmetic until the final approximation, especially for radicals and logarithms.
4. Use the sign of f'' on the interval between a and x to classify over- or underestimation.
5. For measurement error, use $|dy| \approx |f'(x)| |dx|$ as an estimate of maximum propagated error.

Shared Representation / Data

Comparison for $f(x) = x^2 + 1$ and $L(x) = 4x - 3$ at $x = 2$

x	1.98	2.00	2.02	2.10
$f(x)$	4.9204	5.0000	5.0804	5.4100
$L(x)$	4.9200	5.0000	5.0800	5.4000

Worked Example: Square root

Use linearization at $a = 25$ to approximate $\sqrt{25.6}$.

Answer and Reasoning

$f(x) = \sqrt{x}$, $f(25) = 5$, $f'(25) = 1/10$. Thus $L(x) = 5 + (x - 25)/10$ and $\sqrt{25.6} \approx 5.06$.

Worked Example: Cube root

Approximate $\sqrt[3]{8.12}$ using $a = 8$.

Answer and Reasoning

$f(8) = 2$, $f'(8) = 1/12$. Hence $L(8.12) = 2 + 0.12/12 = 2.01$.

Worked Example: Error direction

Is the approximation of $\sqrt{25.6}$ above or below the actual value?

Answer and Reasoning

For $f(x) = \sqrt{x}$, $f''(x) < 0$, so the graph is concave down and the tangent line is above the curve. The approximation is an overestimate.

Worked Example: Measurement error

A circle radius is measured as 10.0 ± 0.03 cm. Estimate maximum area error.

Answer and Reasoning

$A = \pi r^2$, so $dA = 2\pi r dr$. At $r = 10$, $|dA| \approx 20\pi(0.03) = 0.6\pi \text{ cm}^2$.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. Write the linearization of f at $x = a$.

2. For $f(x) = x^2$ at $a = 3$, find $L(x)$.

3. Use $a = 1$ to approximate $\ln(1.04)$.

4. Use $a = 0$ to approximate $e^{0.03}$.

5. Find the linearization of $f(x) = \sin x$ at $a = \pi/6$.

6. Find the tangent approximation for $f(x) = x^{1/5}$ at $a = 32$ and estimate $31.5^{1/5}$.

7. Find a such that the linearization of e^x at a passes through the origin.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. The linearization of f at a is

(A) $f(a) + f'(x)(x - a)$

(B) $f(x) + f'(a)(x - a)$

(C) $f(a) + f'(a)(x - a)$

(D) $f'(a) + f(a)(x - a)$

2. If $f'' > 0$ near a , the tangent approximation is usually

(A) an overestimate

(B) an underestimate

(C) exact everywhere

(D) undefined

3. Using linearization at $x = 9$, $\sqrt{9.08}$ is closest to

(A) 3.0067

(B) 3.0133

(C) 3.0267

(D) 3.0400

Common Errors and AP Precision

- State the quantity, sign, input value, and units in every contextual interpretation.
- In related rates, differentiate before substituting values. In linearization, identify the base point. For L'Hopital's Rule, verify $0/0$ or ∞/∞ first.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

A differentiable function satisfies $f(2) = 5$, $f'(2) = 4$, and $f'' > 0$ on $(2, 2.2)$. Compare $f(2.1)$ with 5.4.

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